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Comprehensive Question Bank: Chapter 8 — Applications of Computer Science

Section A: Multiple-Choice Questions (MCQs)

Topic 8.1a: Artificial Intelligence (AI)

1. Artificial Intelligence (AI) is best described as a branch of computer science that:
A) Connects physical devices to the internet
B) Enables machines to perform tasks that typically require human intelligence
C) Stores data on remote servers
D) Keeps a decentralized, tamper-proof ledger
2. The process by which AI systems learn from experience by being trained on large sets of data is called:
A) Virtualization
B) Machine Learning
C) Actuation
D) Hashing
3. Which technology is loosely inspired by how neurons in the human brain process information?
A) Blockchain
B) Internet of Things
C) Neural Networks
D) Cloud Virtualization
4. NLP, a technology AI uses to simulate how the human brain processes information, stands for:
A) Network Language Programming
B) Natural Language Processing
C) National Language Protocol
D) Neural Logic Processor
5. Which of the following is NOT mentioned as a common everyday application of AI?
A) Voice assistants like Siri and Alexa
B) Chatbots answering customer questions
C) Image recognition identifying faces or medical conditions
D) Cryptographic hashing of supply chain records

6. DeepMind's AI, as mentioned in the chapter, is used to help predict which disease from 3D eye scans?
- A) Diabetes
 - B) Glaucoma**
 - C) Cancer
 - D) Asthma
7. What does "the more data the system processes" generally result in, according to how AI systems learn?
- A) Slower processing speed only
 - B) More accurate results over time**
 - C) Reduced need for algorithms
 - D) Complete elimination of bias
8. Which of the following is a challenge of AI mentioned in the chapter, rather than a benefit?
- A) Personalized learning in education
 - B) Faster performance of repetitive tasks
 - C) Bias in algorithms leading to unfair outcomes**
 - D) Assisting doctors in diagnosis
9. AI supporting education through systems that adapt to each learner's needs is an example of:
- A) Blockchain-based learning
 - B) Adaptive tutoring / personalized learning**
 - C) IoT-based classroom monitoring
 - D) Cloud virtualization
10. In a scenario where an AI hiring tool was trained mostly on resumes of previously hired men, the most likely resulting problem is:
- A) The system will run slower than expected
 - B) The system may unfairly favor male candidates due to biased training data**
 - C) The system will require more cloud storage
 - D) The system will become immutable
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Topic 8.1b: Internet of Things (IoT)

1. The Internet of Things (IoT) refers to a network of physical devices that can mainly:
- A) Mine cryptocurrency automatically
 - B) Collect, share, and act on data through internet connectivity**
 - C) Provide unlimited free cloud storage
 - D) Encrypt financial transactions only

2. Which historical event is credited in this chapter as an early ancestor of the Internet of Things?
- A) The launch of Amazon Web Services in 2006
 - B) The Carnegie Mellon internet-connected Coke machine in 1982**
 - C) The invention of Bitcoin in 2008
 - D) The release of the first neural network model
3. In a typical IoT system, which component is responsible for physically measuring a real-world condition such as temperature or soil moisture?
- A) Actuator
 - B) Sensor**
 - C) Cloud server
 - D) Blockchain node
4. Which component of an IoT system is responsible for physically carrying out an action, such as opening a valve or switching on a light?
- A) Sensor
 - B) Actuator**
 - C) Hash function
 - D) Virtual machine
5. A microcontroller in an IoT system primarily functions to:
- A) Provide unlimited internet bandwidth
 - B) Read sensor data and decide what action to take**
 - C) Mine new blocks for a blockchain
 - D) Store the company's entire cloud database
6. Which wireless technologies are commonly used to connect IoT devices, according to the chapter?
- A) Only fiber-optic cables
 - B) Wi-Fi, Bluetooth, or cellular networks**
 - C) Only satellite uplinks
 - D) Only Ethernet cables
7. In a smart home example from the chapter, smart curtains are set to close automatically when room temperature rises above:
- A) 25°C
 - B) 30°C
 - C) 35°C**
 - D) 40°C
8. Which of the following best summarizes the three main functions of IoT systems as described in the chapter?
- A) Encryption, mining, and validation

B) Data collection, communication, and automation

C) Rendering, streaming, and compression

D) Hosting, scaling, and billing

9. In healthcare, IoT sensors are described as being useful for:

A) Replacing doctors entirely

B) Monitoring patient conditions such as heart rate or glucose levels in real time

C) Mining cryptocurrency for hospitals

D) Encrypting hospital land records

10. A rural smart-irrigation system that uses a GSM module instead of Wi-Fi to transmit sensor data is primarily addressing which real-world constraint?

A) Excess electricity supply

B) Limited or unreliable internet/Wi-Fi connectivity in rural areas

C) Too much available bandwidth

D) A surplus of computing power

Topic 8.1c: Cloud Computing

1. Cloud computing is best defined as a technology that allows users to:

A) Physically connect devices to sensors

B) Store, manage, and process data on remote servers instead of local computers

C) Create decentralized, tamper-proof ledgers

D) Train neural networks without any data

2. Saving a document on Google Drive or iCloud is an everyday example of using:

A) Blockchain

B) IoT

C) Cloud Computing

D) Edge AI

3. The cloud computing feature that allows users to easily increase or decrease their storage and computing power based on need is called:

A) Immutability

B) Scalability

C) Hashing

D) Actuation

4. Which pricing model is described in the chapter as helping cloud users lower operational costs?

A) Fixed annual licensing only

- B) Pay-as-you-go**
C) One-time lifetime purchase
D) Barter-based exchange
5. Which of the following are given as examples of major cloud computing platforms?
A) Bitcoin and Ethereum
B) Amazon Web Services (AWS) and Microsoft Azure
C) GDPR and COMPAS
D) ESP32 and Arduino
6. The chapter's analogy compares cloud computing to:
A) A public spreadsheet that cannot be edited
B) Renting a fully furnished house with electricity on demand, instead of building your own generator
C) A vending machine connected to the internet
D) A courtroom sentencing algorithm
7. Which benefit of cloud computing allows users to work from multiple devices and locations without manually transferring files?
A) Immutability
B) Accessibility
C) Decentralization
D) Consensus
8. In which year did Amazon launch AWS, turning its internal infrastructure into a publicly rentable service?
A) 2001
B) 2006
C) 2010
D) 2016
9. Virtualization in cloud computing refers to:
A) Encrypting all customer data permanently
B) Splitting one physical server into many separate virtual computers rented to different customers
C) Connecting sensors to microcontrollers
D) Creating an unchangeable chain of records
10. Which of the following is NOT typically supported by cloud computing, according to the chapter?
A) Streaming platforms
B) Online learning systems
C) Collaborative workspaces
D) Physical soil-moisture measurement

Topic 8.1d: Blockchain Technology

1. Blockchain technology is best described as a system that:
 - A) Stores information on a single central server for speed
 - B) Securely stores and shares information across many computers, with each block linked to the previous one**
 - C) Only functions without any internet connection
 - D) Replaces the need for sensors in IoT systems
2. The individual in the chapter's story who is credited with solving the "double-spending problem" and inventing Bitcoin's blockchain in 2008 is known by the pseudonym:
 - A) Alan Turing
 - B) Satoshi Nakamoto**
 - C) Tim Berners-Lee
 - D) Ada Lovelace
3. A cryptographic hash, as used in blockchain, is best described as:
 - A) A password shared between all users
 - B) A unique digital "fingerprint" for data that changes completely if the data changes**
 - C) A type of cloud storage plan
 - D) A wireless communication protocol
4. The term "immutable," as applied to blockchain records, means:
 - A) The data can be edited freely by the original creator
 - B) The data cannot be changed once verified and added to the chain**
 - C) The data is stored only temporarily
 - D) The data is visible to no one
5. Blockchain increases transparency and trust primarily by:
 - A) Storing all data on one government server
 - B) Removing intermediaries and allowing every computer in the network to hold a verified copy of the records**
 - C) Deleting old records automatically
 - D) Requiring a single administrator password
6. In finance, blockchain technology powers which well-known application?
 - A) Cloud storage subscriptions
 - B) Cryptocurrencies such as Bitcoin**
 - C) Voice assistants
 - D) Facial recognition attendance systems
7. The chapter's analogy describes blockchain as being like:
 - A) A private diary only one person can read
 - B) A public Google Sheet that thousands of people copy, where no one can ever delete**

or overwrite old rows

- C) A vending machine connected to a network
 - D) A single locked filing cabinet
8. In supply chain management, blockchain is primarily used to:
- A) Predict future weather patterns
 - B) Track goods from origin to customer and ensure authenticity**
 - C) Replace all cloud servers
 - D) Train machine learning models
9. Why would altering a single confirmed block on a blockchain be extremely difficult in practice?
- A) Because only one computer stores the data
 - B) Because it would require changing that block and every block after it on the majority of network computers simultaneously**
 - C) Because blockchains automatically reject all new blocks
 - D) Because blockchain has no internet connection
10. Which sector is NOT mentioned in the chapter as a use case for blockchain?
- A) Financial transactions
 - B) Supply chain tracking
 - C) Land records and academic credentials
 - D) Real-time voice-to-text translation**
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Topic 8.2: Cultural and Ethical Implications of AI Systems

- 1. Which group in an AI system's stakeholders is primarily responsible for designing and programming the AI models?
 - A) Users
 - B) Developers**
 - C) Governments
 - D) Communities
- 2. Which stakeholder group is primarily focused on preserving social values and cultural norms in relation to AI?
 - A) Developers
 - B) Users
 - C) Communities**
 - D) Investors
- 3. The real-world case discussed in the chapter, where an AI tool used in judicial bail/sentencing decisions was found to discriminate based on demographic bias, is known as:
 - A) GDPR

B) COMPAS

C) AWS

D) NLP

4. Why was the bias in the COMPAS case particularly concerning, according to the chapter's discussion?

A) It made the system run more slowly

B) It reflected and amplified unequal patterns already present in historical data, harming a specific demographic group

C) It increased the cost of cloud storage

D) It had no real-world consequences

5. The ethical principle that requires an AI system's decisions to be explainable to the people affected by them is called:

A) Scalability

B) Transparency

C) Virtualization

D) Immutability

6. The ethical principle that ensures there is a clear answer to "who is responsible if an AI system causes harm?" is:

A) Fairness

B) Accountability

C) Automation

D) Consensus

7. Bias in AI most commonly originates from:

A) The color scheme of the user interface

B) Hidden imbalances or patterns within the training data

C) The speed of the internet connection

D) The brand of the cloud provider

8. Designing AI language models to support diverse languages and dialects, rather than only dominant ones, is primarily an effort to ensure:

A) Faster processing speed

B) Cultural inclusivity and fairness

C) Lower cloud storage costs

D) Stronger encryption

9. The General Data Protection Regulation (GDPR) is a policy framework from which region, focused on data privacy?

A) United States

B) European Union

C) Pakistan

D) United Nations Security Council

10. A conflict of interest between businesses and users, as described in the chapter, most commonly arises over:

A) Choice of programming language

B) Data collection practices, where businesses want more data and users want privacy

C) The color of a mobile app's interface

D) The physical location of cloud servers

11. If a face-recognition system is trained mainly on images of one age group and skin tone, it is likely to:

A) Work perfectly for all users regardless of background

B) Perform poorly for people from other groups not well represented in training data

C) Automatically correct its own bias

D) Stop functioning altogether

12. Inclusivity in AI design means ensuring that technology:

A) Is available only to the wealthiest users

B) Serves all people fairly, regardless of gender, ethnicity, or background

C) Is written only in one global language

D) Focuses exclusively on urban populations

13. In the scenario of an AI system used for student admissions, ethical design requires that the university be able to:

A) Hide its selection criteria from all applicants

B) Explain why a student was selected or rejected, and identify who is responsible for the decision

C) Allow the AI to make the final decision with no human oversight

D) Ignore applicant privacy concerns entirely

14. Which of the following best represents "safety" as an ethical principle in AI systems, using the chapter's healthcare example?

A) Making diagnoses as quickly as possible regardless of accuracy

B) Testing AI systems carefully so they do not give harmful or misleading medical results

C) Reducing the amount of data doctors can access

D) Removing all human doctors from the process

Topic 8.3: Designing Applications for Pakistan Using Emerging Technologies

1. Which of the following is identified in the chapter as a national challenge in Pakistan's agriculture sector?
A) Excess water supply
B) Inefficiency and low productivity
C) Overqualified labor force
D) Too much government transparency
2. According to the chapter, which technology can be used to store and distribute educational content efficiently to address barriers in Pakistan's education sector?
A) Blockchain
B) Cloud Computing
C) Actuators
D) GSM modules only
3. In the case study on Smart Agriculture in Pakistan, IoT sensors are primarily used to:
A) Predict long-term national economic trends
B) Collect real-time data on soil moisture, temperature, and nutrients
C) Replace all farm labor
D) Encrypt land ownership documents
4. In the Smart Agriculture case study, AI is primarily used for:
A) Watering crops manually
B) Predictive weather analysis to help farmers plan and protect crops
C) Mining cryptocurrency
D) Physically harvesting crops
5. In the Smart Agriculture case study, blockchain's specific role is to:
A) Predict rainfall patterns
B) Add transparency to the supply chain by recording each stage of crop production and distribution
C) Store soil moisture readings directly from sensors
D) Provide electricity to rural sensors
6. A student designing a technology solution for a national challenge in Pakistan should, according to the chapter, begin by:
A) Choosing as many advanced technologies as possible
B) Identifying a specific, real-world problem before selecting the technology
C) Copying an existing international system exactly
D) Avoiding any research into the people affected
7. Which of the following best reflects the ethical and cultural considerations emphasized for designing applications in Pakistan?
A) Applications should use only English and ignore regional languages

- B) Applications must be inclusive, accessible, and respectful of local customs and traditions**
- C) Applications should avoid all forms of user data protection
- D) Applications should prioritize cost over cultural relevance
8. A smart irrigation system that combines IoT sensors, AI prediction, and Cloud storage is an example of:
- A) Blockchain-only design
- B) Technology integration**
- C) A single-technology solution
- D) Manual farming practice
9. Which of the following is a realistic constraint that should shape smart agriculture system design in rural Pakistan, as highlighted in class discussion?
- A) Unlimited high-speed internet everywhere
- B) Intermittent electricity and limited high-speed connectivity**
- C) Excess computing budgets for every farmer
- D) A complete absence of mobile phones
10. Sending irrigation alerts via SMS instead of a smartphone app in a rural technology deployment primarily addresses which concern?
- A) Data encryption strength
- B) Accessibility for farmers without smartphones or reliable data plans**
- C) Blockchain consensus speed
- D) Cloud server scalability
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Topic 8.4: Policies, Regulations, and Responsible Innovation

1. AI governance refers to:
- A) The hardware components of an AI system
- B) The rules, policies, and guidelines used to ensure AI systems are developed and used responsibly**
- C) A type of neural network architecture
- D) A cloud pricing model
2. Responsible innovation, as defined in the chapter, means creating technology that:
- A) Focuses only on maximizing short-term profit
- B) Solves real problems while considering ethics, safety, culture, fairness, and long-term social impact**
- C) Avoids all forms of regulation
- D) Ignores community input entirely

3. "Privacy-by-Design" refers to the practice of:
- A) Adding privacy protections only after a data breach occurs
 - B) Building data protection into a system's architecture from the very start**
 - C) Removing all data protection to improve system speed
 - D) Allowing unrestricted data collection by default
4. Which professionals, according to the chapter, help ensure AI-based technology projects comply with laws and regulations?
- A) Sociologists
 - B) Legal experts**
 - C) Sensor manufacturers
 - D) Cloud hardware technicians
5. Which professionals study the social impact of technology within interdisciplinary teams building responsible technology?
- A) Legal experts
 - B) Sociologists**
 - C) Microcontroller engineers
 - D) Cloud administrators
6. Why does the chapter emphasize that "no single engineer knows everything about AI, Cloud, IoT, and Blockchain simultaneously"?
- A) To discourage students from studying computer science
 - B) To highlight that responsible technology development requires interdisciplinary collaboration**
 - C) To argue that regulation is unnecessary
 - D) To suggest that only governments should build technology
7. The chapter's AWS launch story (2006) is used to illustrate which point about policy and responsibility?
- A) That cloud computing has no risks at all
 - B) That easy access to powerful computing infrastructure creates both opportunity and the need for responsible governance**
 - C) That AWS was the first blockchain platform
 - D) That cloud computing eliminates the need for data privacy laws
8. Adopting global AI governance standards, such as GDPR-inspired frameworks, can benefit Pakistan by:
- A) Eliminating the need for any local innovation
 - B) Aligning technological growth with ethical standards while building public trust and global partnerships**
 - C) Preventing any use of AI within the country
 - D) Removing the need for interdisciplinary collaboration

9. "Safety-by-Design" primarily involves:
- A) Fixing safety issues only after a system has failed publicly
 - B) Anticipating how a system could fail or be misused and building safeguards before deployment**
 - C) Avoiding any testing before launch
 - D) Excluding domain experts from the design process
10. According to the chapter, technological solutions should represent the diversity of the populations they serve mainly to ensure:
- A) Faster algorithm performance only
 - B) That systems work fairly for everyone, not just a selected group**
 - C) Lower development costs only
 - D) Reduced need for legal compliance
11. A responsible national rollout of a new technology system, as described in the chapter, should include which step before scaling nationally?
- A) Skipping testing to save time
 - B) Pilot testing in a small number of locations to gather feedback and fix problems**
 - C) Immediate nationwide deployment with no monitoring
 - D) Avoiding all interdisciplinary review
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Section B: Short Answer Questions (SQs)

Topic 8.1a: Artificial Intelligence (AI)

1. Define Artificial Intelligence (AI) in your own words.
2. What is Machine Learning, and how does it relate to AI?
3. Differentiate between Neural Networks (NN) and Natural Language Processing (NLP).
4. List three common everyday applications of AI mentioned in the chapter.
5. Explain briefly how DeepMind's AI is used in eye disease prediction.
6. State two benefits and two challenges of AI in society.
7. Explain briefly why bias can appear in an AI hiring system trained on historical data.
8. Predict what might happen to an AI model's accuracy if it is trained on very little data.

Topic 8.1b: Internet of Things (IoT)

1. Define the Internet of Things (IoT).
2. Differentiate between a sensor and an actuator in an IoT system.
3. Briefly explain the role of a microcontroller within an IoT system.
4. Map out, in simple steps, how data moves from a soil sensor to a farmer's mobile alert.

5. State the three main functions of IoT systems described in the chapter.
6. Identify two wireless technologies commonly used to connect IoT devices.
7. Explain briefly how IoT is used in healthcare, according to the chapter.
8. Explain briefly why the Carnegie Mellon Coke machine is considered an early example of IoT.

Topic 8.1c: Cloud Computing

1. Define Cloud Computing.
2. Differentiate between local computer storage and cloud-based storage.
3. Explain briefly what "scalability" means in the context of cloud computing.
4. State two advantages of cloud computing mentioned in the chapter.
5. Identify two real-world examples of cloud computing platforms.
6. Explain briefly the "pay-as-you-go" pricing model.
7. Explain briefly what virtualization means and why it matters for cloud computing.

Topic 8.1d: Blockchain Technology

1. Define Blockchain technology.
2. Differentiate between a centralized database and a blockchain ledger.
3. Explain briefly what a cryptographic hash is and why it matters in blockchain.
4. State two real-world applications of blockchain mentioned in the chapter.
5. Explain briefly why blockchain records are described as "immutable."
6. Identify the historical problem that blockchain (via Bitcoin) was originally designed to solve.

Topic 8.2: Cultural and Ethical Implications of AI Systems

1. Define the term "stakeholder" in the context of AI systems.
2. Differentiate between the roles of developers and users as AI stakeholders.
3. List the four core ethical principles in AI design mentioned in the chapter.
4. Explain briefly why cultural awareness matters when designing AI systems.
5. Explain briefly how bias entered the COMPAS algorithm.
6. State one example of a conflict of interest between a business and its users regarding AI.
7. Explain briefly the purpose of the GDPR framework.
8. Identify two questions you would ask when auditing an AI system for fairness.

Topic 8.3: Designing Applications for Pakistan Using Emerging Technologies

1. List three national challenges in Pakistan that could be addressed using emerging technologies.

2. Explain briefly the role of IoT sensors in the Smart Agriculture case study.
3. Explain briefly the role of AI in the Smart Agriculture case study.
4. Explain briefly the role of Blockchain in the Smart Agriculture case study.
5. Differentiate between designing a technology solution "problem-first" versus "technology-first."
6. Identify two real-world constraints (e.g., connectivity, electricity) that should shape technology design in rural Pakistan.
7. State why cultural and ethical considerations are important when designing applications for local communities.

Topic 8.4: Policies, Regulations, and Responsible Innovation

1. Define AI Governance.
 2. Differentiate between Privacy-by-Design and Safety-by-Design.
 3. List three types of professionals who might collaborate on a responsible technology team, besides computer scientists.
 4. Explain briefly why interdisciplinary collaboration is important in technology development.
 5. State two steps involved in a responsible national technology rollout.
 6. Explain briefly how the story of AWS's 2006 launch relates to the need for technology governance.
 7. Identify one benefit Pakistan could gain from adopting global AI governance standards.
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Section C: Long / Extensive Questions (LQs)

Topic 8.1: Understanding Emerging Technologies

1. Explain how Artificial Intelligence (AI), the Internet of Things (IoT), Cloud Computing, and Blockchain are transforming modern industries.
 - a) Define each of the four technologies in your own words.
 - b) Give one real-world example of each technology in use.
 - c) Explain how these four technologies could work together within a single integrated system.
2. Discuss the benefits and challenges of Artificial Intelligence (AI) in society.
 - a) Describe at least three benefits of AI, with examples from the chapter.
 - b) Describe at least three challenges or risks associated with AI.
 - c) Explain how ethical principles can help ensure AI is used fairly and responsibly.

3. Describe how IoT devices collect and communicate data for automation.
 - a) Explain the roles of sensors, microcontrollers, and actuators within an IoT system.
 - b) Trace, step-by-step, how data flows from a sensor to a final automated action.
 - c) Discuss how IoT is improving at least two of the following sectors: healthcare, agriculture, smart homes.
 4. Explain how Cloud Computing supports scalability, accessibility, and collaboration.
 - a) Define scalability, accessibility, and cost-effectiveness in the context of cloud computing.
 - b) Compare cloud platforms such as Google Drive, AWS, and Microsoft Azure in terms of their typical use cases.
 - c) Discuss one risk or limitation that organizations should consider when relying on cloud computing.
 5. What is Blockchain technology? Discuss how its decentralized nature improves transparency and security.
 - a) Explain, step by step, how a new transaction becomes a permanently recorded "block."
 - b) Discuss how blockchain improves transparency and security in a financial application (e.g., cryptocurrency).
 - c) Discuss how blockchain improves transparency and security in a non-financial application (e.g., supply chain tracking or land records).
 6. Construct a full system architecture diagram and written explanation for an integrated smart-city application that uses all four emerging technologies (AI, IoT, Cloud, Blockchain) together.
 - a) Identify the specific real-world problem your smart-city application solves.
 - b) Diagram and explain the flow of data from data collection through to final action.
 - c) Justify why each of the four technologies was necessary, and what would be lost if one were removed.
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Topic 8.2: Cultural and Ethical Implications of AI Systems

1. Analyze the role of different stakeholders in the design, deployment, and regulation of AI systems.
 - a) Identify and describe the four stakeholder groups discussed in the chapter.
 - b) Explain how the interests of these stakeholder groups can come into conflict, using a specific example.
 - c) Propose one strategy for balancing these competing interests fairly.
2. Discuss the ethical and cultural considerations involved in designing AI systems.
 - a) Explain the four core ethical principles of AI (fairness, accountability, transparency, safety) with an example of each.
 - b) Discuss how cultural diversity can affect the design and reception of an AI system in

different regions.

- c) Evaluate how developers can ensure inclusivity and respect for local values when designing AI for a diverse population like Pakistan's.
 3. Evaluate the real-world case of algorithmic bias described in the chapter (the COMPAS system).
 - a) Explain what the COMPAS algorithm was designed to do, and what bias was discovered in its outcomes.
 - b) Analyze how such bias could have entered the system despite no one explicitly programming discrimination.
 - c) Propose at least two concrete steps developers could take to detect and reduce similar bias in a new AI system.
 4. Design an ethical evaluation process for a new AI system, using a school-based facial recognition attendance system as your example.
 - a) Identify at least three potential ethical, privacy, or cultural risks of this system.
 - b) Propose specific design changes or safeguards to address each risk you identified.
 - c) Discuss who should be held accountable if the system produces an unfair or harmful outcome, and why.
 5. Discuss the role of policy and legal frameworks in governing the ethical use of AI.
 - a) Explain the purpose and key protections offered by the GDPR.
 - b) Discuss why a country like Pakistan might choose to develop its own national AI and data-protection policies rather than simply adopting another country's framework unchanged.
 - c) Evaluate the potential consequences, for both citizens and businesses, of having no AI governance policy at all.
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Topic 8.3: Designing Applications for Pakistan Using Emerging Technologies

1. Identify key national challenges in Pakistan and describe how emerging technologies can provide effective and innovative solutions.
 - a) Identify at least four national challenges across different sectors (e.g., agriculture, healthcare, education, governance).
 - b) For each challenge identified, explain which emerging technology (or combination) could realistically address it, and how.
 - c) Discuss one risk or limitation that could prevent successful deployment of your proposed solutions in Pakistan.
2. Construct a complete end-to-end system design for the Smart Agriculture case study described in the chapter.
 - a) Diagram the full data flow, from IoT soil sensors through Cloud storage, AI prediction, and (optionally) Blockchain record-keeping, to a final farmer-facing action.

- b) Explain the specific role each technology plays at each stage of the system.
 - c) Discuss how this system could be redesigned to function reliably in an area with intermittent electricity and limited internet connectivity.
3. Design an original technology-based application (different from the Smart Agriculture case study) that addresses a specific national challenge in Pakistan.
- a) Clearly define the specific problem your application solves and who it serves.
 - b) Explain which emerging technologies your application uses, and justify why each one is necessary.
 - c) Discuss the ethical and cultural considerations you would need to address before deploying your application nationally.
4. Evaluate the importance of technology integration when designing solutions for national-scale problems.
- a) Explain, with examples, what "technology integration" means in the context of this chapter.
 - b) Compare a single-technology solution to a multi-technology integrated solution for the same problem, and discuss the advantages of integration.
 - c) Discuss the added complexity or risks that come with integrating multiple technologies into a single system.
5. Discuss the importance of ethical and cultural considerations when designing technology solutions for diverse communities within Pakistan.
- a) Explain why a technology solution that works well in one country or region might fail in another.
 - b) Discuss at least three specific cultural or accessibility factors that should shape the design of a Pakistan-focused application.
 - c) Propose a process a student design team could follow to ensure their application remains inclusive and locally appropriate.
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Topic 8.4: Policies, Regulations, and Responsible Innovation

1. Describe the importance of policies, regulations, and interdisciplinary collaboration in promoting responsible innovation.
- a) Explain what is meant by "AI governance" and why it is necessary.
 - b) Describe the roles of at least three different types of professionals (e.g., legal experts, sociologists, ethicists, domain experts) in building responsible technology.
 - c) Discuss how Pakistan can benefit from adopting global AI governance standards while still addressing its own local needs.
2. Analyze the concepts of Privacy-by-Design and Safety-by-Design as approaches to responsible technology development.
- a) Define both concepts and explain the key difference between them.

- b) Using the Smart Agriculture system (or another system from this chapter) as an example, explain how Privacy-by-Design and Safety-by-Design principles could be applied.
 - c) Evaluate what could go wrong if a technology system is deployed without applying either of these principles.
3. Design a responsible national rollout plan for a new AI-powered healthcare-triage application.
- a) Outline the key steps your rollout plan would include, from initial impact assessment to nationwide deployment.
 - b) Identify at least two potential risks (privacy, safety, fairness, or accessibility) and explain how your plan addresses each.
 - c) Justify why pilot testing in a small number of locations should occur before a full national launch.
4. Evaluate the relationship between rapid technological innovation and the need for regulation, using the story of Amazon Web Services (AWS) as a starting point.
- a) Summarize how AWS's 2006 launch changed access to computing power for individuals and small businesses.
 - b) Discuss both the opportunities and the risks created by this kind of easy, large-scale access to powerful computing infrastructure.
 - c) Propose at least two policy safeguards that could help ensure such infrastructure is used responsibly.
5. Discuss how interdisciplinary collaboration can improve the fairness and inclusivity of a national-scale technology system.
- a) Explain why computer scientists alone are typically insufficient to evaluate the full impact of a large technology deployment.
 - b) Describe, with examples, how each of the following could contribute to a responsible technology project: legal experts, sociologists, ethicists, and domain experts (e.g., doctors, agronomists).
 - c) Evaluate how the absence of interdisciplinary collaboration could lead to a technology system that is technically excellent but socially harmful.
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End of Comprehensive Question Bank — Chapter 8: Applications of Computer Science.